Design Methodology of a Multiplying DAC for a 10-bit, 100 MS/s Pipeline ADC in IHP 130 nm CMOS

Giorgis Ezequiel

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Contents

1 Introduction **2**

2 System-Level Specifications and Derived MDAC Targets **2**
2.1 ADC specifications . . . 2
2.2 Derived amplifier requirements . . . 2
2.2.1 Gain . . . 2
2.2.2 Unity-Gain Bandwidth and Phase Margin . . . 3
2.2.3 Thermal Noise and Sampling Capcitor . . . 3
2.2.4 Requirements summary . . . 4
2.3 Corners definition . . . 4

3 Amplifier design **4**
3.1 Topology Selection . . . 4
3.2 Gain . . . 5
3.3 Gain enchancement . . . 5
3.4 Frequency response . . . 5
3.5 Slew Rate and Tail Current . . . 6
3.6 Noise . . . 6
3.7 Bias Network . . . 6
3.8 Switched capacitor common mode feedback . . . 6
3.9 Complete Schematics . . . 7

4 Simulations results **7**
4.1 Open Loop . . . 7
4.2 Input common mode range . . . 8
4.3 Close Loop . . . 9
4.3.1 FFT . . . 9

Abstract

This paper presents the complete design flow of a MDAC stage for a medium-resolution, high-speed pipeline ADC implemented in IHP 130 nm CMOS usign open source tools. The target converter operates at 1.5V, 100 MS/s with 10-bit resolution and 500 mV_{pp} differential output swing, with robust operation across process, voltage, and temperature (PVT) corners. Starting from system-level requirements, we derive quantitative MDAC constraints on thermal noise, closed-loop settling, static gain accuracy, slew-rate, unity-gain bandwidth (UGBW), and phase margin for a given capacitive load.

1 Introduction

Pipeline Analog-to-Digital Converters (ADCs) are widely employed in modern high-speed, medium-to-high resolution applications due to their favorable trade-off between sampling rate, power efficiency, and achievable accuracy.

This type of ADCs chieve high throughput by distributing quantization across multiple stages, each composed of a sub-ADC, sub-DAC, residue amplifier, and sample-and-hold function. The MDAC is the central block in each stage because it both reconstructs the coarse quantization level and amplifies the residue for the next stage. For an ideal 1.5-bit/stage implementation, the residue transfer is

$$V_{res} = 2V_{in} - D \cdot V_{ref}, \tag{1}$$

where $D \in \{-1, 0, +1\}$ is the digital decision and the gain factor of 2 is realized by capacitor ratio and closed-loop OTA action.

2 System-Level Specifications and Derived MDAC Targets

2.1 ADC specifications

The intended converter targets are summarized in Table 1.

Table 1: Target system specifications

Parameter	Target
Resolution	10 bits
Sampling rate f_s	100 MS/s
Output swing	500 mV_{pp} differential
Load capacitance C_L	1.25 pF
Supply voltage V_{DD}	1.5 V
Common-mode level V_{CM}	0.75 V
Technology	IHP 130 nm CMOS

2.2 Derived amplifier requirements

2.2.1 Gain

For a closed-loop residue gain of 2, finite OTA gain A_0 introduces fractional gain error roughly

$$\varepsilon_g \approx \frac{1}{\beta A_0}, \tag{2}$$

where β is the feedback factor (typically near 0.5 in a gain-of-2 MDAC). Requiring residue gain error to be significantly below 0.5 LSB referred to full scale leads to

$$\varepsilon_g \lesssim \frac{1}{2^{N+1}} \quad \therefore \quad A_0 \gtrsim \frac{2^{N+1}}{\beta}. \tag{3}$$

For $N = 10$ and $\beta = 0.5$, this implies $A_0 \gtrsim 4096$ (≈ 72 dB). With digital calibration and redundancy, relaxed targets are possible; nevertheless 70–75 dB open-loop gain is a robust design objective.

2.2.2 Unity-Gain Bandwidth and Phase Margin

At 100 MS/s, the clock period is 10 ns. With two non-overlapping phases, available settling time in amplification phase is approximately

$$T_{\phi 2,eff} \approx 4\text{–}5 \text{ ns} \tag{4}$$

after subtracting switch dead-time and comparator latching overhead.

For first-order settling,

$$|\varepsilon_{dyn}| \approx e^{-t/\tau}, \qquad \tau \approx \frac{1}{2\pi\beta\,\mathrm{UGBW}}. \tag{5}$$

Imposing $\varepsilon_{dyn} < 1/2^{N+1}$ at $t = T_{\phi 2,eff}$ yields

$$\mathrm{UGBW} \gtrsim \frac{\ln(2^{N+1})}{2\pi\beta T_{\phi 2,eff}}. \tag{6}$$

For $N = 10$, $\beta = 0.5$, and $T_{\phi 2,eff} = 4.5$ ns, UGBW is around 540 MHz.

2.2.3 Thermal Noise and Sampling Capcitor

The thermal noise introduced by the sampling switch is shaped by the RC network formed by the switch resistance R and the sampling capacitor C_s, which behaves as a first-order low-pass filter with transfer function:

$$A(j\omega) = \frac{1}{1 + j\omega RC_s} \tag{7}$$

The total sampled noise is obtained by integrating the filtered noise power spectral density over frequency:

$$\overline{e_n^2} = \frac{1}{2\pi}\int_0^\infty \overline{e_R^2} \cdot |A(j\omega)|^2\, d\omega \tag{8}$$

Substituting the thermal noise of the resistor:

$$\overline{e_R^2} = 4kTR \tag{9}$$

$$\overline{e_n^2} = \frac{1}{2\pi}\int_0^\infty \frac{4kTR}{1 + (\omega RC_s)^2}\, d\omega \tag{10}$$

Solving the integral:

$$\overline{e_n^2} = \frac{1}{2\pi} \cdot \frac{4kTR}{RC_s} \cdot [\arctan(\omega RC_s)]_0^\infty = \frac{kT}{C_s} \tag{11}$$

Thus, the sampled noise depends only on the sampling capacitance:

$$\sigma_n^2 = \frac{kT}{C_s} \tag{12}$$

Imposing the resolution constraint:

$$\sigma_n \le \frac{LSB}{2}, \quad LSB = \frac{V_{FS}}{2^N} \tag{13}$$

$$C_s \geq \frac{kT}{(LSB/2)^2} \tag{14}$$

In practice:

$$C_s \approx \alpha \cdot \frac{kT}{(LSB/2)^2}, \quad \alpha \approx 3\text{–}10 \tag{15}$$

2.2.4 Requirements summary

Table 2: Target OTA specifications

Parameter	Target
Gain A_0	$\geq$ 70 dB
Unity-Gain Bandwidth UGBW	$\geq$ 540 MHz
Phase Marigin PM	$\geq 60°$
Output swing	1 V_{pp} diff.
ENOB	$\geq$ 9 bits
Sampling capacitors	$\approx$ 400 fF
Load capacitance C_L	1.25 pF
Supply voltage V_{DD}	1.5 V
Common-mode level V_{CM}	0.75 V

2.3 Corners definition

To ensure the MDAC meets gain, settling, and stability targets under realistic spread, corner-based verification is grouped into three principal process conditions: slow-slow (SS), typical-typical (TT), and fast-fast (FF). Each process corner is evaluated together with coordinated perturbations of supply/common-mode voltages and OTA bias current, all with ±5% variation around nominal values.

Table 3: Corner grouping with process and ±5% bias/supply variations

Group	Process corner	VDD	V_{CM}	I_{bias}
SS group	SS	1.425 V	0.7125 V	47.5 μA
TT group	TT	1.500 V	0.7500 V	50.0 μA
FF group	FF	1.575 V	0.7875 V	52.5 μA

3 Amplifier design

3.1 Topology Selection

A folded cascode OTA with PMOS input pair is selected to maximize input common-mode range and reduce flicker noise. The PMOS input devices allow operation with input voltages close to ground, which is suitable for switched-capacitor MDAC environments. The folded structure provides high gain while maintaining moderate voltage headroom under low supply conditions.

The signal path consists of a PMOS differential pair with transconductance g_{m1}, whose currents are folded into NMOS cascode devices, producing a high output resistance node.

./FoldedCascode.jpg

Figure 1:

3.2 Gain

For a folded-cascode stage, low-frequency gain can be approximated as

$$A_0 \approx g_{m1} R_{out}, \tag{16}$$

where R_{out} is the small-signal resistance at the output node. With cascoding,

$$R_{out} \approx (g_{m7,8} r_{O7,8} r_{O9,10}) \parallel [g_{m3,4} r_{O3,4} (r_{O5,6} \parallel r_{O1,2})] \,. \tag{17}$$

Hence, gain is increased primarily by: (i) maximizing g_{m1}, and (ii) increasing r_O through longer channel devices.

3.3 Gain enchancement

To meet gain under 130 nm intrinsic gain limits, gain enhancement (gain-boosting) is applied to NMOS cascode transistors. If auxiliary amplifiers provide local gain A_1, then effective output resistance scales approximately as

$$R_{out,eff} \approx R_{out} \, (1 + A_1), \tag{18}$$

Practical implementation must preserve stability; therefore gain-boosting amplifier poles should be placed sufficiently above MDAC closed-loop bandwidth, or compensated to avoid excess phase lag near unity crossover.

The auxiliary amplifier itself is implemented using a folded cascode topology, which provides high gain. Because the ouput swing is relative small, to stabilize its output common-mode voltage, a continuous-time common-mode feedback (CMFB) network based on MOS transistors operating in the triode region is employed. These devices act as voltage-controlled resistors, setting the output common-mode level without introducing significant additional poles.

3.4 Frequency response

The frequency response of the folded cascode OTA is analyzed using the node-pole association method, which allows identifying the dominant poles by associating each high-impedance node

with its corresponding resistance and capacitance. Neglecting gain boosting effects, the response is dominated by three main nodes: the output node, the internal folding node X, and the input node.

The dominant pole is typically located at the output due to the large load capacitance C_L and high output resistance R_{out}:

$$\omega_{p,out} \approx \frac{1}{R_{out}C_L} \tag{19}$$

The internal node X, corresponding to the folded branch (drain of the input pair and source of the cascode devices), introduces a non-dominant pole given by:

$$\omega_{p,X} \approx \frac{1}{R_X C_X} \tag{20}$$

with:

$$R_X \approx \frac{1}{g_{m3,4}}, \quad C_X \approx C_{gs3,4} + C_{sb3,4} + 2C_{gd5,6} + C_{db5,6} \tag{21}$$

Stability requires sufficient separation between $\omega_{p,out}$ and the higher-frequency poles at node X and the input, indicating the need of a high $g_{m3,4}$ to ensure a phase margin greater than 60°.

At the input, the pole is defined by the source resistance and the total input capacitance:

$$\omega_{p,in} \approx \frac{1}{R_{in}C_{in}} \tag{22}$$

where:

$$R_{in} \approx \frac{1}{g_{m1}}, \quad C_{in} \approx C_{gs1} + C_{gd1}(1 + A_v) \tag{23}$$

The Miller multiplication of C_{gd1} can significantly increase the effective input capacitance, although in switched-capacitor applications this effect is partially mitigated by the sampling network.

Overall, the unity-gain bandwidth is primarily determined by:

$$UGBW \approx \frac{g_{m1,2}}{C_L} \tag{24}$$

To reach a UGBW of 540 MHz for a load of $C_L = 1.25$ pF, the $g_{m1,2}$ will be

$$g_{m1,2} = 2\pi 540\,\text{MHz} \times 1.25\,\text{pF} = 4.2\,\text{mS} \tag{25}$$

Considering a $g_m/I_D = 12$

$$I_{D_{M0}} = \frac{4.2\text{mS}}{12} \approx 350\,\mu\text{A} \tag{26}$$

However, as it will be described in the next section, this current might be higher due to the slew-rate requirements.

3.5 Slew Rate and Tail Current

3.6 Noise

3.7 Bias Network

3.8 Switched capacitor common mode feedback

The switched-capacitor common-mode feedback (SC-CMFB) circuit regulates the output common-mode voltage by periodically sensing and correcting it through charge redistribution. During phase ϕ_1, the sampling capacitors acquire the output common-mode level:

$$V_{CM,out} = \frac{V_{O1} + V_{O2}}{2} \tag{27}$$

In phase ϕ_2, the capacitors are reconnected to the reference common-mode voltage $V_{CM,ideal}$, generating an error proportional to:

$$\Delta V_{CM} = V_{CM,out} - V_{CM,ideal} \tag{28}$$

This error is transferred to the control node V_{CMFB}, which adjusts the OTA biasing (e.g., $M_{5,6}$ gate voltage), closing the feedback loop and forcing:

$$V_{CM,out} \rightarrow V_{CM,ideal} \tag{29}$$

The switched-capacitor implementation ensures accurate common-mode sensing and is inherently compatible with discrete-time operation. Proper capacitor matching and non-overlapping clock phases are required to minimize offset, charge injection, and instability in the CMFB loop.

3.9 Complete Schematics

4 Simulations results

4.1 Open Loop

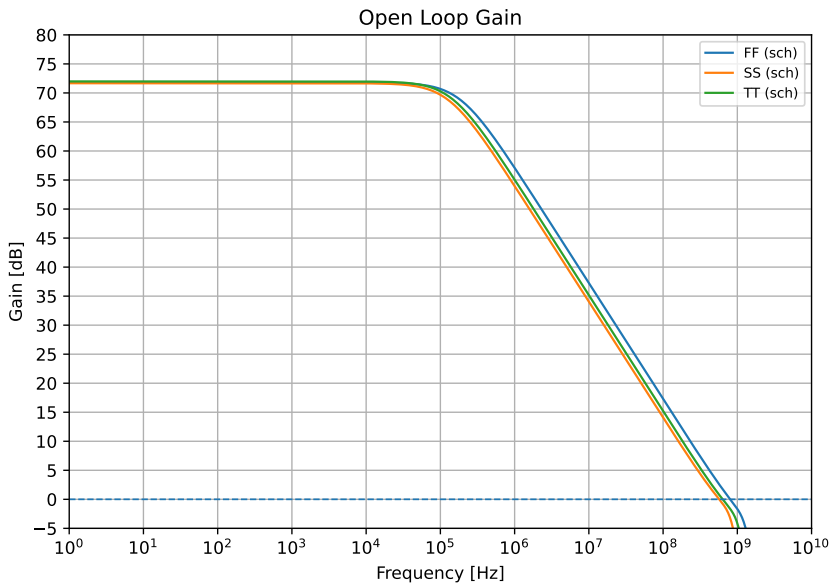

Figure 2: Open Loop Gain

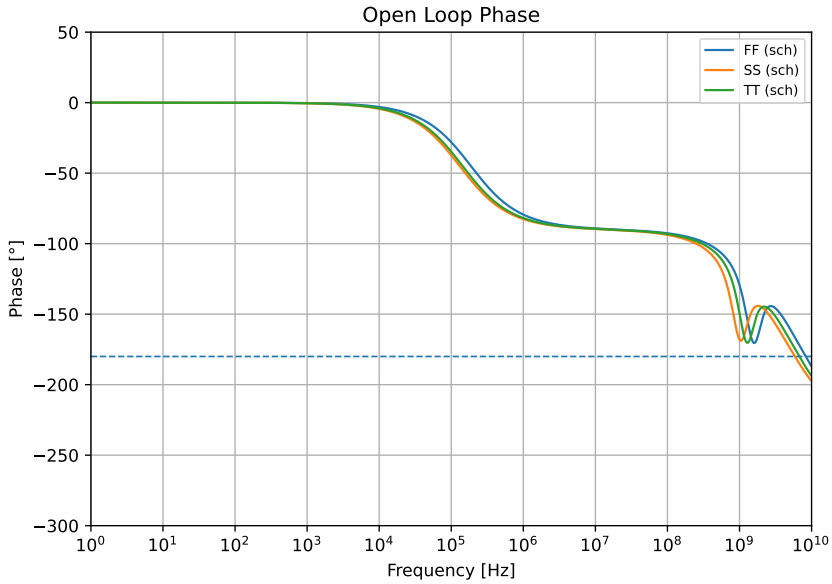

Figure 3: Open Loop Phase

4.2 Input common mode range

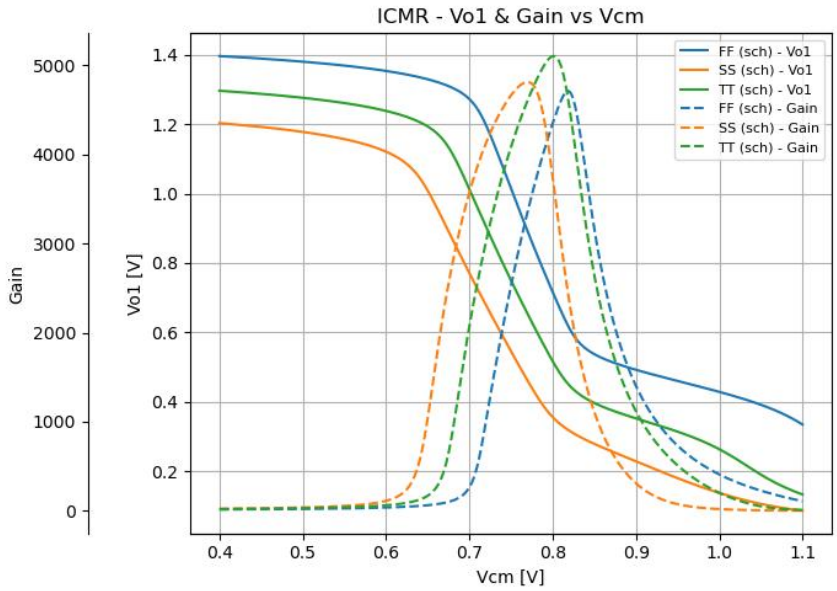

Figure 4:

4.3 Close Loop

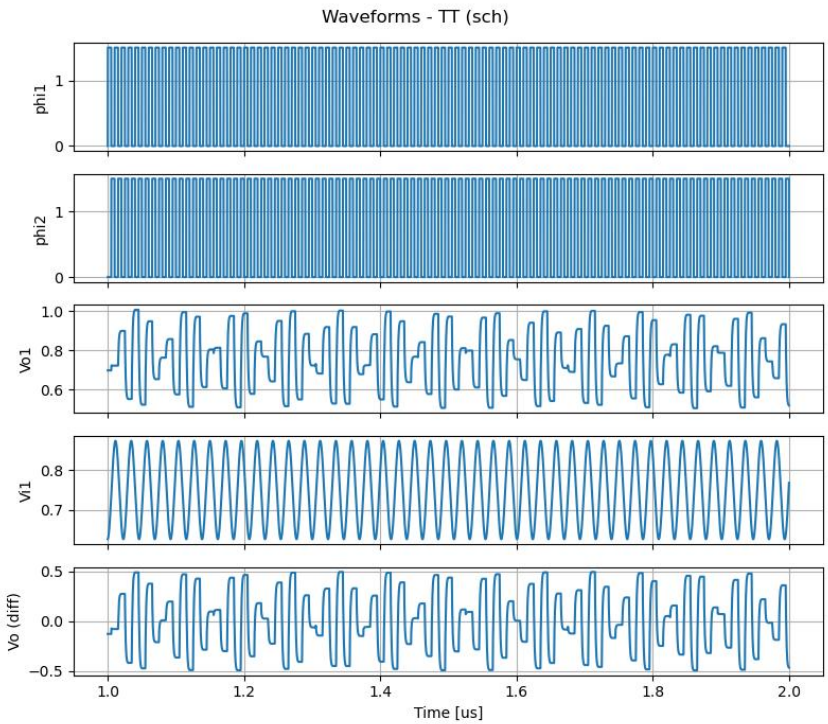

Figure 5:

4.3.1 FFT